Leveraging AI Agents and Multi-Agent Debate To Automate MLB Front Office Decisions and Roster Construction

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Abstract

This project leverages artificial intelligence Agents and Multi-Agent Debate (MAD) to evaluate Major League Baseball players and construct a twenty-six player MLB roster. The AI Agents evaluate each player based on statistics and grades them according to a five-tool scale, eventually resulting in a single numerical rating. MAD is then employed alongside optimization techniques to select thirteen hitters and thirteen pitchers, maximizing team performance according to agent consensus, while staying within a specified salary cap. The operating hypothesis is that AI agents can effectively evaluate Major League Baseball players based on performance statistics, even with large and complex datasets. By applying Multi-Agent Debate (MAD) and optimization techniques, a competitive team can be constructed that maximizes overall performance while adhering to a specified salary cap, mirroring real-world decision-making processes in sports management.

Introduction

In Major League Baseball, players are evaluated through a variety of methods. While traditional statistics such as batting average, home runs, runs batted in, and stolen bases remain important, the rise of advanced sabermetrics has transformed scouting and player assessment. With the growing prevalence of artificial intelligence, this process is poised to evolve even further. AI introduces technological capabilities that reduce human bias, automate traditionally manual processes, and enable more objective and efficient evaluations. By training models on large and complex datasets, AI can comprehensively assess player abilities, providing Major League Baseball organizations with valuable insights for informed decision-making. Automating aspects of the evaluation process could significantly expand the scale and depth of analysis achievable by a single individual, ultimately helping organizations build teams more strategically and tailor roster construction to specific attributes or performance metrics.

This project includes data on all hitters and pitchers in Major League Baseball from the past five seasons (2020-2024). There are 24 different statistics chosen for hitters and 30 for pitchers. These statistics range from being more traditional (Average, Earned Run Average, On-Base Percentage), to more advanced (Wins Above Replacement, Fielding, Independent Pitching). Each players average annual value (AAV) is also included in the dataset. This project utilizes two different AI models, ChatGPT-4 and SuperGrok for analysis to evaluate each player and assign each of them a single numerical rating. ChatGPT-4 is a large language model developed by OpenAI that can understand and generate human-like text, enabling users to ask questions, generate code, summarize information, and perform complex reasoning tasks. SuperGrok is an AI-powered tool designed specifically for data analysis and interpretation, capable of extracting insights from large datasets and identifying patterns through advanced machine learning techniques. Together, these methods enable more efficient, scalable, and insightful analysis of complex information, supporting decision-making processes that would otherwise require significant time and expertise.

Methodology

In order to begin the process, a dataset of hitters and pitchers needed to be created to be utilized by the AI Agents. Through using linear regression, 24 different statistics were chosen for hitters and 30 for pitchers. These statistics were taken from Baseball Reference and Fangraphs, which both serve as statistical databases for any player in the history of professional baseball, along with a wide range of sabermetrics used to further analyze player performance. In order to find data on player average annual value (AAV) and team payrolls, Sportrac was used which contains data of team payrolls and player contracts in professional sports.

The next step was to utilize prompt engineering technique with the AI Agents to evaluate each hitter and pitcher. The AI Agents were tasked with evaluating each player according to a five-tool scale (Hitting, Hitting for power, Fielding, Throwing, Speed) and supplying an inclusive overall rating. These ratings were normally distributed and spanned from 1-100, allowing for a wide range of player ratings to be created. For players with multiple seasons played within the time period of the dataset (2020-2024), the AI agents considered all previous seasons within the dataset, as well as the present season, with the more recent seasons being weighed heavier in the overall rating.

A multi-agent debate system is then used to optimize the selection of a twenty-six player MLB roster within a specified salary cap by leveraging five specialized agents for hitters and five separate agents for pitchers, each prioritizing distinct metrics. These agents are aimed to take the place of executives within a team's front office with varying perspectives and preferences for evaluation. For hitters the agents include:

- WAR Agent (prioritizing higher WAR)
- Durability Agent (prioritizing more games played and consistently performing hitters)
- Moneyball Agent (value via performance vs. AAV)
- Traditional Agent - (prioritizing traditional statistics such as AVG, HR, and RBI)
- PowerAgent - (prioritizing more extra-base hits e.g. 2B, HR, SLG)

Each agent independently selects 13 players, assigning confidence scores, which are aggregated into a score matrix during the debate phase. The system then uses the PuLP library, a tool for linear programming (LP) and integer programming problems, to maximize total score while ensuring positional requirements and salary cap compliance. Voting details and player stats are output, providing a transparent, balanced roster selection that integrates diverse performance perspectives.

This process is done with pitchers as well with the following agents:

- FIP Agent (prioritizing low FIP)
- WAR Agent (prioritizing higher WAR)
- Strikeout Agent (prioritizing SO9 and SO/BB)
- Traditional Agent (prioritizing W, L, SO, ERA, WHIP)
- Moneyball Agent (value via performance vs. AAV)

Each agent selects 13 pitchers with confidence scores. These selections are aggregated, creating a score matrix by summing confidence scores. Linear programming via PuLP, maximizes the total score, ensuring 5 starters (GS ≥ 10), 8 relievers (GS < 10), and salary cap compliance. The final output details each pitcher's stats, role, voting confidence, and rationale, ensuring a balanced, cost-effective staff.

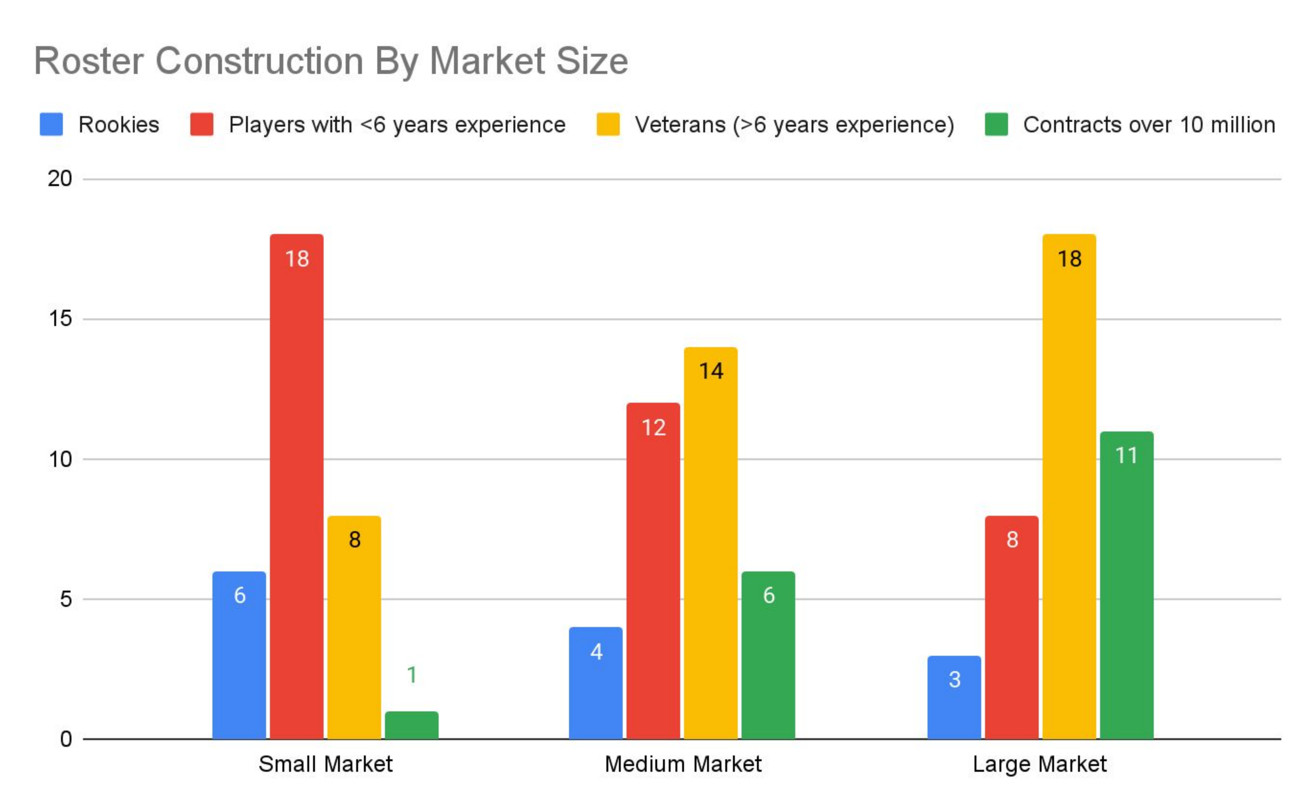
Results

Through using K-means clustering and identifying three clear market sizes of Major League Baseball teams, a team with the payroll of a small market team (e.g. 70 million), a medium market team (e.g. 170 million), and a large market team (e.g. 330 million) were used as case studies. Each team was allowed to spend half of their total payroll on hitters and the other half on pitchers. Using the multi-agent debate system and PuLP library, the player ratings previously calculated were able to be used to optimize a set of thirteen hitters and thirteen pitchers while adhering to positional constraints and a maximum salary cap.

The small market team (total payroll: $70 million) had the lowest overall WAR among the three rosters but was still composed mostly of above-average players. Only one player—a starting pitcher—earned more than $10 million annually. This team relied heavily on younger players who had not yet reached free agency and were likely being paid below their market value. The limited budget also restricted spending on older, more established talent, with only eight veteran players on the roster.

The medium market team (total payroll: $170 million) generated significantly more total WAR than the small market team and featured six players with average annual values (AAVs) over $10 million. While nearly half the team had fewer than six years of experience, the majority of the roster consisted of veteran players. The increased budget allowed for greater investment in proven talent, as demonstrated by the six high-value contracts.

The large market team (total payroll: $330 million) posted the highest total WAR, though only slightly more than the medium market team. Despite its substantial payroll, the team still relied on younger players who outperformed their AAVs, accounting for nearly one-third of the roster. However, the team clearly invested heavily in established talent across all positions of need, with eleven players earning over $10 million annually.

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Player: José Ramírez
WAR: 23.30
BA: 0.278
HR: 145
RBI: 473
OPS+: 142
AAV: $14411711.22
Positions: *5/D
Votes:
  - High WAR Agent: Confidence = 0.70
  - Clutch Agent: Confidence = 0.84
  - Bang for Buck Agent: Not selected
  - Traditional Agent: Confidence = 0.85
  - Power Agent: Confidence = 0.76
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Player: Paul Skenes
Role: Starter
WAR: 29.50
ERA: 1.96
FIP: 2.44
SO/9: 11.50
SO/BB: 5.31
W-L: 11-3
WHIP: 0.947
AAV: $2712955.00
Votes:
  - FIP Agent: Confidence = 0.82
  - WAR Agent: Confidence = 1.00
  - Strikeout Agent: Not selected
  - Traditional Agent: Confidence = 0.93
  - Bang for Buck Agent: Not selected
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Conclusion

By utilizing a multi-agent debate (MAD) system and the PuLP library, AI agents were able to construct full twenty-six player MLB rosters based on diverse evaluation perspectives and statistical preferences. Each roster was built within defined positional constraints and under a specified salary cap. At lower salary caps, teams relied more heavily on rookies and younger players who had not yet reached free agency and were therefore undervalued in terms of AAV. As the salary cap increased, rosters began to include more established veterans with higher salaries, while still prioritizing high-performing younger players. This continued reliance on younger, cost-effective talent helped narrow the performance gap between the medium and large market teams, despite their significant payroll differences.

Limitations and Future Work

This project is ultimately limited by the scope of the available data. Future extensions could incorporate multiple seasons of player statistics and a broader range of metrics, such as biomechanical data and advanced sabermetrics, to enhance the accuracy and depth of player evaluations. Expanding the multi-agent debate (MAD) system to include additional AI agents could also introduce more diverse evaluation strategies, enriching the decision-making process.

Another valuable extension would be to tailor the model to specific MLB teams, using it to address particular roster needs. Since real-world teams cannot acquire any player at will, this approach would improve practicality by focusing on optimizing transactions within the constraints of an existing roster and the current free agent pool. This would also allow for multi-agent debate to include agents that reflect the preferences of specific teams, thus making the roster selection process more individualized.

While automating the evaluation and roster construction process, this project obviously overlooks the human factor of the decision-making process of MLB front offices, which is still valued by teams. Integrating AI-driven analysis with human expertise could offer a more balanced approach, combining the efficiency and objectivity of automation with the intuition and contextual understanding of human evaluators.

Annotated Code and References